

■ CURRENT POSITION 5th-semester B.Tech CSE student with hands-on exposure to DSA, Python, Data Analysis, Machine Learning, Generative AI and practical project work. Current edge: technical learning + college leadership + public project documentation.	■ TARGET GOAL Primary: Secure a strong AI/ML, Data Science or ML Engineering internship and build toward placement-ready roles. Long-term: Develop production-oriented AI/ML capability with a strong portfolio and professional network.
--	--

■ SKILL GAP ANALYSIS			
AREA	CURRENT BASE	GAP TO CLOSE	PRIORITY
Programming + DSA	C/C++/Python; DSA practice	Patterns, complexity, timed problem solving	HIGH
Data Science	Regression, classification, clustering, metrics	Statistics, feature engineering, validation, storytelling	HIGH
ML Engineering	Project-level ML exposure	Git, APIs, Docker, deployment, testing, MLOps basics	VERY HIGH
Generative AI	Prompting + AI exploration	RAG, embeddings, tool use, evaluation, agents	HIGH
Portfolio + Interview	Kaggle/GitHub + challenge work	2–3 flagship projects, README quality, ML/DS interview prep	VERY HIGH

■ RECOMMENDED LEARNING PLAN			
NOW → 2 MONTHS		3 → 4 MONTHS	5 → 6 MONTHS
DSA: arrays, strings, binary search, linked lists, trees, DP basics. DS: statistics, EDA, feature engineering, model validation. Tools: Git/GitHub, SQL, NumPy, Pandas, scikit-learn.		ML: ensembles, tuning, pipelines, imbalance, explainability. Engineering: FastAPI/Flask, Docker, testing, deployment. GenAI: embeddings, RAG, tool calling.	Interview: ML/DS concepts + SQL + DSA timed sets. Portfolio: polish 2–3 flagship projects. AI: build one useful end-to-end AI application and document trade-offs.

■ SUGGESTED PROJECTS	■ NETWORKING STRATEGY
1. End-to-End Student Performance ML App — training → evaluation → API/UI → deployment. 2. RAG Knowledge Assistant — documents → embeddings → retrieval → grounded answers → evaluation. 3. AI Career/Resume Analyzer — resume parsing, skill-gap analysis and role matching with transparent scoring.	Build in public: post one useful learning/project update weekly. Target people: ML engineers, data scientists, recruiters, alumni and founders. Outreach: 5 thoughtful messages/week; ask about learning paths or projects, not just openings. Proof: send a concise GitHub project + 2-line value summary.

■ MONTHLY MILESTONES		
MONTH	FOCUS	DELIVERABLE
SEP 2026	Foundation	Lock DSA routine; strengthen Python/SQL; organize GitHub; finish current AI prompting challenge.
OCT	Core DS/ML	Statistics + EDA + feature engineering; one polished classical ML project.
NOV	ML Depth	Model tuning, pipelines, explainability; start deployment fundamentals.
DEC	GenAI Build	Learn embeddings/RAG/tool use; build a small working GenAI app.
JAN 2027	Interview Mode	Timed DSA + SQL; ML/DS interview notes; resume + GitHub portfolio polish.
FEB	GATE + Portfolio	GATE CSE milestone; maintain interview practice; document flagship project.
MAR	Applications	Targeted internship applications; alumni/recruiter outreach; mock interviews.

■ IMMEDIATE NEXT ACTIONS — NEXT 7 DAYS		
1. Choose one flagship ML project and define the problem, dataset, metric and deployment target. 2. Set a daily 60–90 min DSA block + 45–60 min ML/AI block. 3. Clean GitHub READMEs and pin your strongest projects. 4. Publish one concise learning post and connect with 5 relevant professionals/alumni. 5. Track progress weekly: DSA solved • concepts learned • project commits • applications • conversations.		